MILITARY SPECIFICATION SHEET

LOUDSPEAKER, SHIPBOARD ANNOUNCING SYSTEMS ENCASED, 70.7 VOLT, 80 VOLT-AMPERES

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the loudspeaker described herein shall consist of this specification and the latest issue of MIL-L-24223.

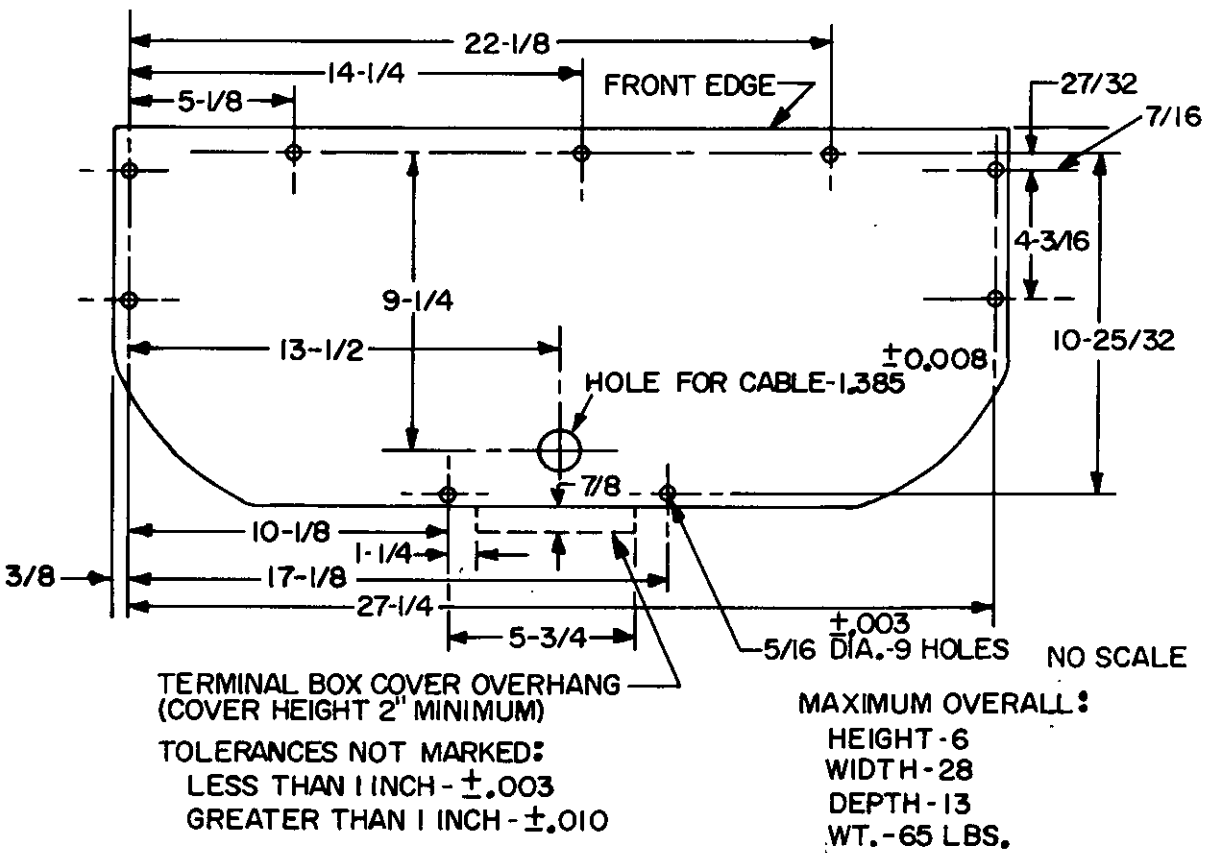


FIGURE 1. Mounting dimensions and cable entrance for LS-657( )/SIC.

AMSC N/A FSC 5965

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REQUIREMENTS

PHYSICAL

Dimensions and weight: See figure 1.

Cable entrance: See figure 1. Cable shall be installed using stuffing tubes in accordance with MIL-S-19622/1, size 2. Stuffing tubes are not required unless otherwise specified.

Enclosure: Mounting base shall be aluminum sheet 6061-T6 in accordance with QQ-A-250/11. Enclosure shall be aluminum sheet 5052-H32 in accordance with QQ-A-250/8. Speaker enclosure shall be spraytight in accordance with MIL-STD-108. Wire connections shall be closed lug on a threaded terminal with locknut and slip over rubber sleeves to protect the electrical connections from water, ice and deposits. Rear drainage holes are to be provided for internal drainage. Exterior surfaces shall drain when level. An aluminum alloy grill, not less than 0.125 inch thick with openings not larger than 9/32 inch shall cover the speaker front. Application is shipboard, exterior, unsheltered.

Input enclosure: Transformer and wiring terminal block shall be in a separate water tight enclosure in accordance with MIL-STD-108. Wiring from the transformer to speaker drivers shall be ceramic insulated (encased) conductors through the enclosure wall or brass stuffing tubes. Application is shipboard, exterior, unsheltered.

Speaker drivers: Speaker driver assemblies shall not exceed five.

PERFORMANCE

Testing: See MIL-L-24223.

Rated continuous sinusoidal single frequency: For any single frequency input from 320 through 3200 Hz, the rated input shall be less than 80.5 rms volt-amperes. For sound pressure outputs specified in table I herein. The input power shall be at least 87 percent of the volt-amperes value.

Peak input: Peak input shall be at least two times the maximum volt ampere input specified in table I herein when tested as specified in MIL-L-24223.

Voice coil impedance: 4 $\pm$ 0.4 ohms.

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Sound pressure output: Minimum rms sound pressure output in dB re 20 micropascals at rated input shall be as specified in table I, when measured in accordance with 4.5.4.4.

TABLE I. Minimum on axis sound pressure (rms dBp).

Nomenclature	Distance to test MIC (feet)	Volt ampere input (max)	Warble bands-Hz					
			320 to 500	500 to 800	800 to 1250	1250 to 2000	2000 to 3200	3200 to 5000
LS 657( )/SIC	10	80	111	110	116	120	116	--

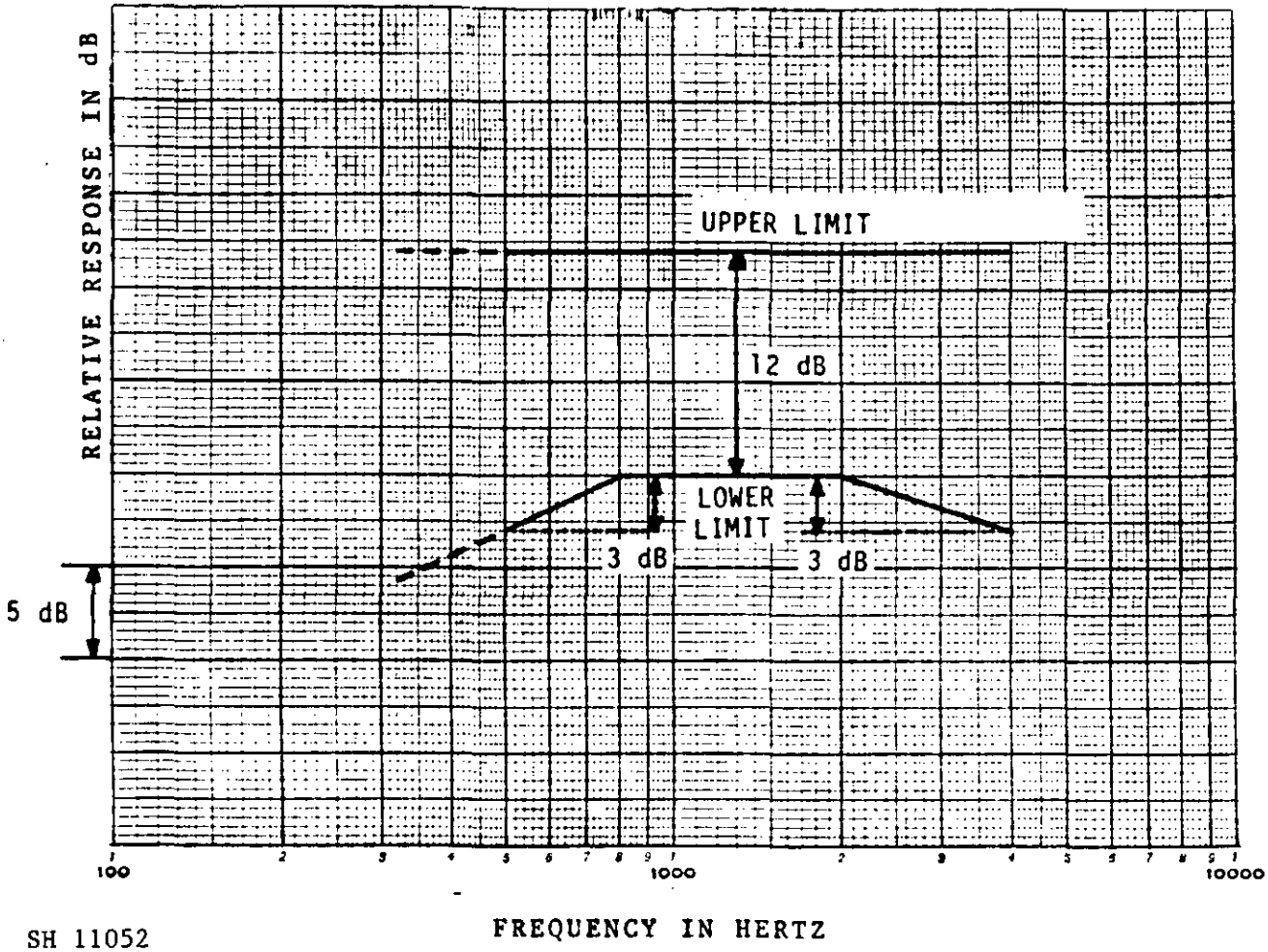
Coverage angle: At least 30 degrees in horizontal and 60 degrees in vertical, from 320 to 2000 Hz.

Sound power: Output level re 10^{-12} watts with 1 watt input shall be at least 106 dB (20 milliwatts) in the 500 to 3200 Hz band.

Frequency: Response shall be within the figure 2 limits over the band shown.

Volume control: None required.

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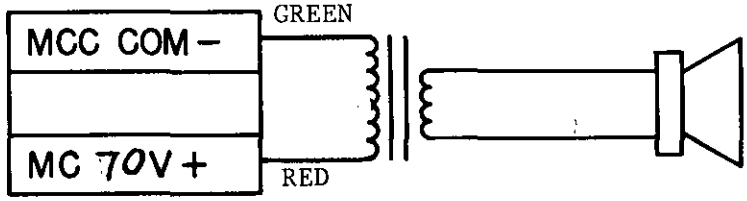


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FIGURE 2. Frequency response limits.

Fundamental resonance: Minimum two speaker elements in the assembly shall be measured and reported.

Marking: Marking shall be as specified in MIL-L-24223/1 and figure 3 herein.



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FIGURE 3. Wiring diagram and terminal designations.

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ENVIRONMENTAL

Requirements and inspection: As specified in MIL-L-24223 and as follows:

Temperature: Required ranges in accordance with MIL-E-16400 for exposed unsheltered application.

Icing: Required.

Gun blast: In accordance with MIL-E-16400. Energized with a 10 to 15 volt 400 Hz signal during test. Post gun blast performance testing shall include verification tests of MIL-L-24223, table VI.

TESTING: See MIL-L-24223.

Mounting: Complete assembly mounted on a 4 foot by 8 foot by 3/4 inch sheet of plywood.

Group C inspection: Group C inspection shall be performed on every 301st to 303rd assemblies that are deliverable. Group C tested assemblies may be refurbished for delivery provided all speaker drivers are replaced.

Preparing activity:
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